

ONLINE FOOD DELIVERY ANALYSIS

Data-Driven Business Insights | Project Documentation

Prepared for project submission | Project: Online Food Delivery Analysis

1. Executive Summary

This project analyzes online food delivery order data to understand customer ordering behavior, revenue patterns, delivery performance, restaurant/cuisine trends, and order cancellations. The workflow combines Python and Pandas for data preparation and exploratory data analysis (EDA), MySQL for structured querying, and Power BI for interactive business reporting.

The working dataset contains 100,000 order records. The analysis includes data quality checks, missing-value treatment, validation of ratings and business rules, feature engineering, visual exploration, SQL-based analysis, and a Power BI dashboard.

2. Problem Statement

Online food delivery platforms generate transactional data about customers, orders, restaurants, payments, and deliveries. The business needs to turn this raw data into reliable insights that can support operational planning and customer experience improvements.

Project objectives: understand customer ordering behavior; track revenue and profit-related metrics; identify delivery and cancellation patterns; compare cities and cuisines; and present key performance indicators through a dashboard.

3. Objectives

- Prepare a clean, analysis-ready dataset from noisy real-world data.
- Explore order value, delivery time, distance, ratings, and cancellation reasons.
- Analyze weekday/weekend and monthly patterns.
- Use SQL queries to answer business questions.
- Build a Power BI dashboard with KPIs and interactive filtering.
- Communicate findings and practical recommendations.

4. Tools and Technologies

Python: data processing and analysis.

Pandas and NumPy: tabular transformations and numerical operations.

Matplotlib and Seaborn: charts and correlation heatmap.

MySQL: relational data storage and analytical SQL queries.

Power BI: interactive dashboard and KPI visualization.

Jupyter Notebook: development environment.

5. Dataset Overview

The project dataset contains 100,000 rows and initially 32 columns in the working notebook. The project brief describes the core dataset as covering customer details, order information, restaurant attributes, delivery performance, and financial metrics.

Representative fields used in analysis include Customer_Age, Order_Date, Order_Day, City, Cuisine_Type, Delivery_Time_Min, Distance_km, Order_Value, Discount_Applied, Final_Amount, Delivery_Rating, Restaurant_Rating, Profit_Margin, and cancellation-related fields.

Note: The exact final schema should be confirmed against the submitted cleaned CSV/database table, because preprocessing and feature engineering add or modify columns.

6. Methodology / Workflow

Step 1 — Data understanding: inspect shape, columns, data types, sample records, summary statistics, and unique values.

Step 2 — Data quality assessment: identify missing values, invalid ratings, inconsistent date/day values, and logically inconsistent records.

Step 3 — Data cleaning: treat missing categorical values, validate ratings, and set ratings for cancelled orders to missing where a customer delivery rating is not applicable.

Step 4 — Feature engineering: derive day type, profit margin percentage, delivery performance categories, rating categories, and age groups.

Step 5 — EDA: use distribution charts, category comparisons, scatter plots, monthly/weekly summaries, and a correlation heatmap.

Step 6 — SQL analysis: store/query structured data and answer business questions using aggregations, grouping, sorting, and filtering.

Step 7 — Power BI: present KPIs, trends, and comparisons in an interactive dashboard.

Step 8 — Documentation: record methodology, findings, limitations, and recommendations.

7. Data Cleaning and Preprocessing

Missing City values: 16,726 records were missing City before treatment. These were filled with the label “Unknown” to preserve rows without inventing a city.

Missing Cuisine_Type values: 16,885 records were missing Cuisine_Type before treatment. These were filled with “Unknown”.

Missing Order_Date: 1,014 records had no date. Date-based analysis should exclude these rows or report them separately. The notebook adjusted Day_Type handling so missing dates did not create false mismatches with the supplied Order_Day.

Ratings: ratings were validated against the expected scale. Ratings for cancelled orders were set to NaN because a delivery rating is not applicable to an order that was not completed. After this treatment, 70,916 non-null Delivery_Rating values remained.

Outlier check — Delivery_Time_Min: Q1 = 45, Q3 = 210, IQR = 165; upper IQR bound = 457.5 minutes. The notebook reported no values above this upper bound.

Outlier check — Order_Value: Q1 = 673, Q3 = 3,494, IQR = 2,821; upper IQR bound = 7,725.5. The notebook reported no values above this upper bound.

Important: An IQR bound is a statistical flag, not automatic proof that a record is erroneous. The notebook’s reported result is that no observations exceeded the stated upper bounds.

8. Feature Engineering

Day_Type: categorized records as Weekday or Weekend. Counts: 71,370 Weekday and 28,630 Weekend.

Profit_Margin_Percentage: calculated as $\text{Profit_Margin} \times 100$. For example, 0.13 corresponds to 13% when Profit_Margin is stored as a decimal fraction.

Delivery performance: Delivery_Time_Min was binned into Fast (0–30), Moderate (30–60), Slow (60–120), and Very Slow (over 120) using the notebook's configured bins.

Rating categories and customer age groups: derived categorical fields were added to support segment-level analysis.

Peak-hour field: the notebook notes that Peak_Hour is a binary indicator, not an actual clock-time value. It should be interpreted as a flag only.

9. Exploratory Data Analysis

9.1 Order value and delivery-time distributions: histograms were used to inspect the distribution and spread of order values and delivery times.

9.2 City-wise orders: before filling missing values, the city counts were Hyderabad 16,884; Bangalore 16,732; Delhi 16,695; Mumbai 16,493; and Chennai 16,470. These counts exclude the missing-city records and should not be treated as the full distribution after Unknown was added.

9.3 Cuisine-wise orders: before filling missing values, counts were Indian 16,685; Arabian 16,658; Chinese 16,651; Mexican 16,602; and Italian 16,519. These exclude missing cuisine records.

9.4 Distance vs delivery time: the observed correlation was approximately 0.003, indicating almost no linear correlation in this dataset. This does not prove that distance has no relationship with delivery time; other factors and non-linear relationships may matter.

9.5 Cancellation reasons: Late Delivery = 3,059; Customer Cancelled = 2,993; Restaurant Issue = 2,979. These are the categories shown in the notebook's value-count output.

9.6 Weekday vs weekend: Weekday — 71,370 orders, revenue 62,040,654, average order value 1,967.05. Weekend — 28,630 orders, revenue 24,842,012, average order value 1,946.41.

9.7 Correlation heatmap: most displayed numeric correlations were close to zero. Order_Value and Final_Amount had a correlation of about 1.00; this is likely influenced by the way final amount is calculated. Discount_Applied and Final_Amount had a correlation of approximately –0.07. Correlation does not establish causation.

10. Key Findings

- The working dataset contains 100,000 order records.
- Cancelled orders total 15,036, representing 15.04% of all orders; delivered orders total 84,964.
- Total revenue reported in the project analysis is ■133,597,720. Confirm the exact revenue definition used in the SQL/Power BI measure before presenting it as gross or net revenue.
- Missing data is material: City 16,726; Cuisine_Type 16,885; Order_Date 1,014; and delivery-time values missing for 5,030 records. The notebook notes that missing delivery times were associated with cancelled orders.
- Weekdays account for more orders and revenue than weekends in the available data. Average order value is slightly higher on weekdays in the notebook's aggregation.
- The three leading cancellation reasons shown are Late Delivery, Customer Cancelled, and Restaurant Issue, with similar counts.

- Distance and delivery time have near-zero linear correlation in this dataset; delivery performance should be investigated alongside traffic, restaurant preparation, time of day, and other available variables.

11. SQL and Database Analysis

The project uses MySQL to store the food-order data and execute business-focused queries. The SQL analysis covers customer spending, age group/order value, weekday versus weekend behavior, monthly revenue, discount/profit relationships, city and cuisine performance, delivery time by city, distance/delay, ratings, restaurant performance, peak-hour demand, payment preferences, and cancellation reasons.

Use the actual submitted SQL script as the source of truth for query names, joins, filters, and metric definitions. Include the SQL file in the submission package so the analysis is reproducible.

12. Power BI Dashboard

The Power BI report is intended to provide an interactive overview of order and business performance. Core KPIs include Total Orders, Total Revenue, Average Order Value, Average Delivery Time, Cancellation Rate, and Average Delivery Rating.

Recommended slicers/filters include city, cuisine, date range, and order status. Visuals should make it easy to compare demand, revenue, delivery performance, and cancellations.

Ensure every KPI is tied to a clearly defined measure and that cancelled orders are handled consistently in delivery-time and rating calculations.

13. Conclusion

This project demonstrates an end-to-end data analytics workflow: data understanding, cleaning, feature engineering, EDA, SQL analysis, and Power BI reporting. The results identify cancellation patterns, data-quality gaps, weekday/weekend differences, and relationships among order and delivery variables. The dashboard and documented SQL queries provide a foundation for monitoring performance and supporting evidence-based operational decisions.